# Log-concavity of length counts for long-step lattice paths

A proof of MathDB problem 405093

Principia Math In Progress

September 12, 2026

**Task 1: solve the original problem.** A step is $(u, 0)$ or $(0, u)$, where $u$ is a positive integer. For an integer $d \geq 4$, let $N_\ell$ count paths of $\ell$ steps from $(2, 1)$ to either $(d-1, d)$ or $(d, d-1)$. We prove Conjecture 6 in the problem of Martina Juhnke recorded in [1, p. 150], including its stronger log-concavity assertion.

**Theorem 1.** *Put $D = 2d - 4$. The numbers $N_\ell$ are positive for $2 \leq \ell \leq D$, zero outside that range, and strictly log-concave on their support. In fact, for $3 \leq \ell < D$,*

$$N_\ell^2 \geq \frac{D - \ell + 1}{D - \ell} N_{\ell-1} N_{\ell+1} > N_{\ell-1} N_{\ell+1}. \tag{1}$$

*In particular, the sequence is unimodal.*

## 1 Counting by compositions

For positive integers $a, b$, let $P_{a,b,\ell}$ count paths from $(0, 0)$ to $(a, b)$ with $\ell$ steps. A path with $h$ horizontal and $v$ vertical steps has $h + v = \ell$. Its step directions can be ordered in $\binom{\ell}{h}$ ways, and its positive step lengths are two compositions, of $a$ into $h$ parts and of $b$ into $v$ parts. Consequently

$$P_{a,b,\ell} = \sum_{\substack{h+v=\ell \\ h,v \geq 1}} \binom{\ell}{h} \binom{a-1}{h-1} \binom{b-1}{v-1}. \tag{2}$$

For $a \geq 1$, define

$$A_a(t) = \sum_{j=0}^{a-1} \binom{a-1}{j} \frac{t^j}{(j+1)!}.$$

Equation (2) gives the exponential generating polynomial

$$\sum_\ell P_{a,b,\ell} \frac{t^\ell}{\ell!} = t^2 A_a(t) A_b(t). \tag{3}$$

Translation of the two endpoints in the question, with $n = d - 2$, therefore yields

$$\sum_\ell N_\ell \frac{t^\ell}{\ell!} = t^2 \big( A_{n-1}(t) A_{n+1}(t) + A_n(t)^2 \big). \tag{4}$$

## 2 An interlacing polynomial family

**Lemma 2.** *For $a \geq 1$, the polynomial $A_a$ has $a - 1$ simple negative zeros. For $a \geq 2$, the zeros of $A_{a-1}$ strictly interlace those of $A_a$.*

*Proof.* Coefficient comparison in the definition gives

$$A_1 = 1, \quad A_2 = 1 + \frac{t}{2}, \qquad (a+1)A_{a+1} = (2a+t)A_a - (a-1)A_{a-1} \quad (a \geq 2). \tag{5}$$

For completeness, after multiplication by $(j+1)!$, the coefficient identity in (5) is

$$(a+1)\binom{a}{j} = 2a\binom{a-1}{j} + (j+1)\binom{a-1}{j-1} - (a-1)\binom{a-2}{j},$$

with the usual zero conventions. Also $A_a(0) = 1$ and its leading coefficient is $1/a!$.

Induct on $a$. The base pair $A_1, A_2$ is immediate. Let $r_1 < \cdots < r_{a-1} < 0$ be the zeros of $A_a$. At each such zero, (5) gives

$$A_{a+1}(r_i) = -\frac{a-1}{a+1} A_{a-1}(r_i). \tag{6}$$

By the induction hypothesis these values are nonzero and alternate in sign, yielding a zero of $A_{a+1}$ in every $(r_i, r_{i+1})$. At $r_{a-1}$ the right side of (6) is negative, whereas $A_{a+1}(0) = 1$, yielding a zero in $(r_{a-1}, 0)$. At $r_1$ its sign is $(-1)^{a-1}$, opposite to the sign $(-1)^a$ of $A_{a+1}$ near negative infinity. There is thus also a zero to the left of $r_1$. These are $a$ distinct negative zeros of a degree-$a$ polynomial, which proves simplicity and strict interlacing. □

**Lemma 3.** *For $n \geq 2$ and $\alpha, \beta > 0$, the polynomial*

$$Q_n(t) = \alpha A_{n-1}(t) A_{n+1}(t) + \beta A_n(t)^2$$

*has $2n-2$ simple negative zeros. For $\alpha, \beta \geq 0$, not both zero, all its zeros are negative, possibly with multiplicities.*

*Proof.* At each zero $r_i$ of $A_n$, (5) gives

$$Q_n(r_i) = -\alpha \frac{n-1}{n+1} A_{n-1}(r_i)^2 < 0.$$

At each zero $s_i$ of $A_{n-1}$, $Q_n(s_i) = \beta A_n(s_i)^2 > 0$. The strict interlacing is $r_1 < s_1 < r_2 < \cdots < s_{n-2} < r_{n-1} < 0$. Moreover $Q_n(0) = \alpha + \beta > 0$, and $Q_n$ is positive near negative infinity because its degree $2n-2$ is even and its leading coefficient is positive. There are therefore $2n-2$ disjoint sign-change intervals, proving the first assertion. If one weight vanishes, the claim follows directly from Lemma 2. □

# 3 Returning from exponential to ordinary counts

We include the needed form of Newton's inequalities.

**Lemma 4.** *Suppose $F(t) = \sum_{\ell=0}^{D} c_\ell t^\ell/\ell!$ has nonnegative coefficients and only nonpositive real zeros. If $c_{\ell-1}c_{\ell+1} > 0$ and $1 \leq \ell < D$, then*

$$c_\ell^2 \geq \frac{D-\ell+1}{D-\ell}\, c_{\ell-1}c_{\ell+1}. \tag{7}$$

*Proof.* By Rolle's theorem $R = F^{(\ell-1)}$ has only nonpositive real zeros. Its degree is $m = D-\ell+1 \geq 2$ and its positive constant $R(0) = c_{\ell-1}$ makes all its zeros strictly negative. Write them as $-\rho_1, \ldots, -\rho_m$, with $\rho_i > 0$. Logarithmic differentiation and Cauchy–Schwarz give

$$\frac{R''(0)}{R(0)} = \left(\sum_i \frac{1}{\rho_i}\right)^2 - \sum_i \frac{1}{\rho_i^2} \leq \frac{m-1}{m} \left(\frac{R'(0)}{R(0)}\right)^2.$$

Using $R'(0) = c_\ell$ and $R''(0) = c_{\ell+1}$ gives (7). □

*Proof of Theorem 1.* The right side of (4) has degree $D = 2n$, a double zero at zero, and otherwise simple negative zeros by Lemma 3. All its coefficients of degrees $2, \ldots, D$ are positive. Lemma 4 proves (1). The positive consecutive ratios $N_{\ell+1}/N_\ell$ consequently decrease, which proves unimodality. □

The same proof applies to a nonzero nonnegative weighted sum of the two endpoint counts, since Lemma 3 permits arbitrary nonnegative $\alpha, \beta$. Equation (3) and Lemmas 2 and 4 also show that the length counts to *any* fixed endpoint $(a, b)$, $a, b \geq 1$, are strictly log-concave on their support. Under the usual convention allowing the empty path, the smaller cases in the question are immediate: for $d = 2$ the ordinary counting polynomial is 1, and for $d = 3$ it is $t + 3t^2$.

# 4 Scope and verification

The paths used here are precisely those in the definition preceding Conjecture 6 in the published workshop report. The report defines the coherent-path sequence separately, imposing diagonal avoidance and compatibility for Conjecture 5. The present proof resolves Conjecture 6. The author's current related papers were checked for later claims; see `literature-audit.md`.

The proof is algebraic and independent of computation. `verify_counting.py` independently checks (2) against a path dynamic program for 169 rectangles, verifies 39 recurrence identities, and checks (4) and the quantitative inequalities for 39 dimensions. Its exact results are saved in `exact-checks.json`.

# 5 Follow-up tasks

**Task 2: impose diagonal avoidance.** Prove log-concavity of the corresponding length counts when paths may cross the diagonal but may not have a vertex on it.

**Task 3: resolve the coherent-path conjecture.** Prove the log-concavity assertion for the coherent-path counts in Conjecture 5 of the same workshop report, including its compatibility restriction.

**Task 4: classify wider endpoint mixtures.** For a fixed $r$, determine the nonnegative weights $\lambda_0, \ldots, \lambda_r$ for which $\sum_{j=0}^{r} \lambda_j A_{n-j}(t) A_{n+j}(t)$ has only real zeros for every $n \geq r + 1$.

**Task 5: determine the typical length.** For a uniformly chosen path counted in Theorem 1, find asymptotic formulas for the mode, mean and variance of its length as $d \to \infty$, and prove a central limit theorem.

# References


[1] G. Chapuy, S. Elizalde and I. Fischer (organizers), *Enumerative Combinatorics*, Oberwolfach Rep. 23 (2026), no. 1, 81–156, published 21 July 2026. Open problems, problem of Martina Juhnke, Conjecture 6, p. 150. doi:10.4171/OWR/2026/2.

[2] G. Poullot, *Vertices of the monotone path polytopes of hypersimplices*, arXiv:2411.14102.

[3] M. Juhnke and G. Poullot, Unimodality of the number of paths per length on polytopes: examples, counterexamples, and a central limit theorem, *Int. Math. Res. Not.* 2026, no. 17, rnag191. doi:10.1093/imrn/rnag191. Current author manuscript: arXiv:2504.20739v3, 11 August 2026.